Zhengzhong Ricky You

San Jose, California | +1 (352) 872-8929 | ricky.you.or@gmail.com
Homepage | Google Scholar | LinkedIn | ORCID | GitHub

Research Interests

- Exact algorithms for large-scale integer and combinatorial optimization, with an emphasis on branch-price-and-cut, decomposition, and extended formulations.
- Learning-augmented optimization, decision-focused learning, and reinforcement learning for combinatorial optimization, together with interfaces between operations research and machine learning.
- Applications of large language models to healthcare operations, including routing, scheduling, and resource allocation under uncertainty and fairness constraints.

Education

- **University of Florida**, M.S. in Industrial and Systems Engineering **2025**
- **Nankai University**, B.S. in Chemistry **2021**

Publications

1. Two-Stage Learning to Branch in Branch-Price-and-Cut Algorithms for Solving Vehicle Routing Problems Exactly.
Zhengzhong Ricky You, Yu Yang, Xinshang Wang, Wotao Yin. *Operations Research*, 2026.
2. RouteOpt: An Open-Source Modular Exact Solver for Vehicle Routing Problems.
Zhengzhong Ricky You, Yu Yang. *INFORMS Journal on Computing*, 2026.
3. Equity-Enforced Path Recommendation in Congested Transit Systems During Disruptions.
Baichuan Mo, **Zhengzhong Ricky You**, Ruimin Li. Major revision at *Transportation Science*.
4. Bulk Service Queueing for Transit Resilience under Short Random Service Suspensions.
Baichuan Mo[†], **Zhengzhong Ricky You**[†], Li Jin, Zuo-Jun Max Shen, Haris N. Koutsopoulos, Jinhua Zhao. Reject and resubmit at *Transportation Science*.
5. Learned Pairwise Deep Dual-Optimal Inequalities for Stabilizing Column Generation.
Zhengzhong Ricky You, Bo Tang, Haoran Liu, Baichuan Mo. Under review at *INFORMS Journal on Computing*.
6. PrecPack: An Efficient Open-Source Exact Solver for Bin Packing with Generalized Precedence Constraints.
Sunkanghong Wang, **Zhengzhong Ricky You**, Roberto Baldacci, Baichuan Mo, Hu Qin, Lijun Wei, Zhou Xu. Under review at *INFORMS Journal on Computing*.
7. KKT Atlas: Learning Low-Dimensional Local Scenario Models for Contextual Stochastic Optimization.
Zhengzhong Ricky You, Baichuan Mo, Yuan Zhou, Jinhua Zhao, Zuo-Jun Max Shen. To be submitted to *Operations Research*.
8. [‡] Debiasing Optimal Values in Stochastic Programming by Averaging Solutions.
Henry Lam, Huajie Qian, Kaizhao Sun, **Zhengzhong Ricky You**. To be submitted to *Operations Research*.
9. Driver Surge Maps for Supply Rebalancing with Downstream Matching.
Yida Xu[†], Baichuan Mo[†], **Zhengzhong Ricky You**[†], Yuan Zhou, Hao Yi Ong, Zuo-Jun Max Shen. To be submitted to *Manufacturing & Service Operations Management*.
10. A Dual-Guided Column-Generation-Based Matheuristic for Integrated Railroad Blocking.
Yida Xu[†], **Zhengzhong Ricky You**[†], Baichuan Mo. To be submitted to *Transportation Science*. Third Prize, 2026 INFORMS RAS Problem Solving Competition.
11. TorchDCM: A Unified PyTorch-Native Package for Discrete Choice Modeling.
Baichuan Mo, **Zhengzhong Ricky You**[*], Xiqun Michael Chen, Ruimin Li.
12. Log-Driven Semantic Tuning for Mixed Integer Programming Solvers.
Zhengzhong Ricky You, Kaizhao Sun, Zeping Min, Wotao Yin, Xinshang Wang.
13. Draft-and-Audit Reinforcement Learning for Optimization Modeling.
Zeping Min, Weihang Xu, **Zhengzhong Ricky You**, Wotao Yin, Xinshang Wang. *ICML 2026*.
14. Towards Routing Foundation Models for Commercial Vehicles.
Bo Tang, Yunfei Ma, **Zhengzhong Ricky You**. *UMFM 2026*, ACM SIGSPATIAL.

[*] Corresponding author. [†] Co-first authors; contributed equally. [‡] Authors listed alphabetically by last name.

Research Experience

- **TikTok Inc.** **2026**
  Machine Learning Engineer Intern, E-commerce Supply Chain & Logistics *San Jose, CA*
  - Researched heuristic algorithms for inventory control in e-commerce supply chains, under the mentorship of Baichuan Mo, Yikang Hua, and Zhou Ye.
  - Developed an LLM-based agent to analyze algorithm outputs and investigate the reasons behind observed results, under the mentorship of Yufei Ruan.
- **MIT-UF-NEU Joint Summer Research Camp** **2026**
  Selected Participant, MIT JTL-Transit Lab program *Remote*
  - Developed **KKT Atlas** for contextual stochastic optimization under the mentorship of Baichuan Mo. Combined local context partitioning, gradient-informed dimension reduction, and joint learning of compact scenario supports and context-dependent probabilities.
  - Analyzed sufficient conditions under which a d-dimensional local uncertainty representation admits $d+1$ shared scenarios. Derived decision-error bounds and quantified the tradeoff between finer localization and estimation error from smaller local samples.
  - Evaluated the method on capacity allocation, portfolio optimization, and network reservation problems. Used 0.8–3.6% of the reference support size and reduced deployment time by 92–98% relative to the full-support comparator, with the lowest or nearly lowest mean risk among the evaluated baselines.
- **University of Kentucky** **2026**
  Visiting Scholar, Department of Statistics *Lexington, KY*
  - Conducted research on online learning algorithms during a visit hosted by Chenglong Ye.
- **DAMO Academy, Alibaba Group** **2025**
  Research Intern *Bellevue, WA*
  - Studied solver parameter tuning with large language models under the mentorship of Xinshang Wang and Wotao Yin. The resulting method, Log-Driven Semantic Tuning, reduced tuning loss by 44% and 17% relative to the strongest tested baseline on easy and intermediate instances, respectively.

Research Software

- **PrecPack** **2026**
  Co-developer, C++ *Open source*
  - Co-developed an exact solver for bin packing with generalized precedence constraints, combining branch-bound-and-remember search, independently checked solutions, and reproducible batch workflows.
- **TorchDCM** **2026**
  Co-developer, Python *Open source*
  - Co-developed a unified PyTorch-native package for discrete choice modeling. The package implements vectorized likelihoods and automatic differentiation across multiple logit model families and provides standard econometric outputs.
- **RouteOpt** **2021**
  Lead Developer, 20k+ LOC in C++ *Open source*
  - Designed and implemented an open-source exact solver for the CVRP and CVRPTW. The solver uses modular branching, cutting-plane, and variable-reduction components, learning-augmented branching, and reproducible experiment scripts.
  - On benchmark instances, RouteOpt nearly doubled the speed of VRPSolver and proved optimality for four previously open instances; one required approximately 1.5 years of single-core time.

Selected Relevant Courses

- **ISE Ph.D. level (GPA: 4.00)**: Discrete Optimization; Linear Programming and Network Optimization; Stochastic Modeling & Analysis; Advanced Topics in Continuous Optimization.
- **Mathematics Ph.D. level (GPA: 4.00)**: Analysis I; Numerical Linear Algebra; Numerical Analysis; Numerical Optimization.
- **CS Ph.D. level (GPA: 4.00)**: Machine Learning.

Conference & Workshop Presentations

- **Finalist presentation**, INFORMS Railway Applications Section (RAS) Problem Solving Competition, INFORMS Annual Meeting, San Francisco, November 2 **2026**
- *L-DDOIs: Learning-Based Deep Dual-Optimal Inequalities for Stabilizing Column Generation*, INFORMS Transportation Science and Logistics Conference, MIT Sloan **2026**
- *Two-Stage Learning to Branch in Branch-Price-and-Cut Algorithms for Solving Vehicle Routing Problems Exactly*, invited talk, UK AI and ML Symposium and Nontechnical Workshop **2025**
- *L-DDOIs: Learning-Based Deep Dual-Optimal Inequalities for Stabilizing Column Generation*, INFORMS Annual Meeting **2025**
- *Accelerating Dynamic Programming via Dual Selection for Column-Generation Frameworks*, INFORMS Optimization Society Conference **2024**
- *RouteOpt: A Scalable Advanced Optimization Tool for VRPs*, INFORMS Annual Meeting **2024**
- *Two-Stage Learning to Branch in Branch-Price-and-Cut for Exact VRP*, INFORMS Annual Meeting **2023**
- *Learning to Branch with Column Generation*, INFORMS Annual Meeting **2022**

Awards & Honors

- **Third Prize**, INFORMS Railway Applications Section (RAS) Problem Solving Competition **2026**
- Harold D. Haldeman, Jr. Graduate Fellowship, University of Florida **2024**
- INFORMS Optimization Society Conference Travel Support **2024**
- Academic Excellence Scholarship, Nankai University **2020**
- Innovation and Entrepreneurship Scholarship, Nankai University **2019**
- First Prize, Tianjin Theoretical Chemistry Competition **2019**
- First Prize, Tianjin Mathematics Competition **2018**

Teaching & Mentoring

- **Sole Instructor**, ESI 3327C Matrix & Numerical Methods in Systems Engineering, University of Florida; taught linear algebra, numerical methods, and introductory optimization to 52 undergraduates **2024**
- **Outreach Mentor**, UF Student Science Training Program; Ken Cheng, Columbia University (computer science), and Shrey Gupta, University of Oxford (mathematics)

Professional Service

Conference reviewer for the *International Conference on Learning Representations (ICLR)*. Journal referee for *INFORMS Journal on Computing*, *INFORMS Journal on Optimization*, *Transportation Science*, *Transportation Research Part B*, *IISE Transactions*, *Computers & Operations Research*, and *OR Spectrum*.

Skills

Methods: branch-price-and-cut, column generation, dual stabilization, cutting planes, learning-augmented optimization, stochastic and robust optimization. **Programming**: C++, Python including PyTorch and reinforcement learning libraries, MATLAB, Shell, and Git. **Software**: PrecPack, RouteOpt, TorchDCM, Gurobi, CPLEX, SCIP, and reproducible experiment pipelines.